MIL-A-23685B(OS)
27 May 1977
SUPERSEDING
MIL-A-23685A(OS)
27 July 1973
and
AMENDMENT 1
29 January 1975

MILITARY SPECIFICATION

ADAPTERS, PALLET, STEEL FOR PROJECTILES, ROCKET HEADS, CARTRIDGE CONTAINERS AND BOMB FINS

* This specification has been approved by the Naval Sea Systems Command (OS), Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

* 1.1 Scope. This specification covers steel wire pallet adapters intended for use in palletizing projectiles, rocket heads, cartridge containers and bomb fins (see 6.1).

* 1.2 Classification. The pallet adapters shall be of the following types as specified (see 6.2).

Type I - Adapter, Pallet MK 11 MOD 1 for 5 inch projectiles and rocket heads
Type II - Adapter, Pallet MK 20 MOD 0 for 6 inch, 47 caliber projectiles
Type III - Adapter, Pallet MK 98 MOD 0 for 8 inch, 55 caliber projectiles
Type IV - Adapter, Base, Pallet MK 11 MOD 2 for 5 inch 54 caliber projectiles
Type V - Adapter, Pallet MK 132 MOD 0 for Container, Cartridge 4.4 inch chaff (RBOC)
Type VI - Adapter, Pallet ADU-383/E for Bomb fin assembly MK 15 and MODS (SNAKEYE)
Type VII - Adapter, Pallet ADU-426/E for bomb fin assembly MK 82 (500 LB. G.P.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization Department, Code 611, Indian Head, MD 20640, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3990

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2. APPLICABLE DOCUMENTS

2.1 The following documentation of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

* SPECIFICATIONS

FEDERAL

QQ-S-781	Strapping Steel Flat and Seals

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage

* DRAWINGS

Naval Sea Systems Command (CODE IDENT 10001 & 53711)

LD 276447 (Dwg. 1341956)	Adapter, Pallet MK 11 MOD 1 (General Arrangement)
LD 483344 (Dwg 1564364)	Adapter, Pallet MK 20 MOD 0 (Assembly)
DL 2642774 (Dwg 2642774)	Adapter, Base, Pallet MK 11 MOD 2 (Assembly)
DL 2643801 (Dwg 2643801)	Adapter, Pallet MK 98 MOD 0 (Assembly)
DL 2645496 (Dwg 2645496)	Adapter, Pallet MK 132 MOD 0 (Assembly)
DWG 1350881	Expendable Pallet

Naval Air Systems Command (CODE IDENT 30003)

DL 752AS100	Adapter, Pallet ADU-383/E
DL 1107AS100	Adapter, Pallet ADU-426/E

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(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other Publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated the issue in effect on date of invitation for bids or request for proposal shall apply.

American National Standards Institute

ANSI Y32.3-1969	Welding Symbols

(Applications for copies should be addressed to American National Standards Institute, 1430 Broadway, New York, New York 10018).

* American Society For Testing and Materials (ASTM)

ASTM A 82	Cold Drawn Steel Wire
ASTM A 123	Zinc (Hot Galvanized) coatings on fabricated products.
ASTM A 185	Welded steel wire fabric for concrete reinforcement
ASTM A 386	Zinc coatings (hot dip) on assembled steel products
ASTM E 380-76	Standards for Metric Practice (IEEE-268-1975 ANSI 2210.1)

(Application for copies of the ASTM standards by serial designation and title should be made to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103)

3. REQUIREMENTS

* 3.1 General Requirements. The pallet adapters shall conform to the applicable LD's or DL's as listed below and all documents listed thereon, this specification and the contract or order.

Type I - LD 276447
Type II - LD 483344
Type III - DL 2643801
Type IV - DL 2642774
Type V - DL 2645496
Type VI - DL 752AS100
Type VII - DL 1107AS100

* 3.1.1 Material. Wire material shall be cold-drawn steel wire conforming to ASTM A 82. All other material shall be as specified in applicable drawings.

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* 3.1.2 Protective Coating. The pallet adapters shall be hot dip galvanized in accordance with ASTM A 123 or ASTM A 386 Class B1 and B2 or otherwise coated in accordance with the applicable drawings.

3.2 Detail Requirements

* 3.2.1 Construction. The construction of the pallet adapters shall be strictly in accordance with the drawings and specifications listed on the applicable LD;s or DL's for each pallet adapter. (see 3.1).

* 3.2.1.1 Metric. When metric size pallet adapters are specified, conversion shall be as specified in ASTM E 380-76.

* 3.2.2 Gages. The supplier shall provide himself with whatever gages are necessary and adequate to ensure that the material to which this specification applies will meet the dimensional requirements shown on the applicable drawings.

* 3.2.3 Conflicting requirements. Conflicting requirements arising between this specification and any specification, publicaton, or drawing shall be decided according to the following order of presentation.

a. Contract
b. This specification
c. Applicable drawing
d. Other specification or publications

Conflicts arising shall be referred in writing to the procuring activity or appointed agent for interpretation, clarification, and resolution (or correction).

* 3.3 Performance requirements and product characteristics. The pallet adapters shall meet the following performance requirements and product characteristics:

* 3.3.1 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease and other foreign matter. All intersections of welded wire fabric shall be welded in accordance with ASTM A 185. All other welding shall be as specified in applicable drawings. The interpretation of welding symbols shall be in accordance with ANSI Y32.3-1969.

3.3.1.1 Weld Machine. All machines used for assembly of the pallet adapters shall be qualified for the combination of particular material thickness prior to being used in production. Qualification shall consist of the production of satisfactory specimens tested in accordance with 4.5.1.

3.3.1.2 Cross Wire Weld Shear. When tested as specified in 4.5.1.1 and 4.5.1.2, the welds shall not fail. Failure of any specimen to meet these requirements shall constitute a failure of the qualification. The welding method or equipment shall be modified and the test repeated.

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3.3.1.3 Butt Weld-Tension. When tested as specified in 4.5.1.3 the welds shall not fail. Failure of any specimens to meet these requirements shall constitute a failure of the qualification. The welding method or equipment shall be modified and the test repeated.

* 3.4 Workmanship. All surfaces and edges shall be free from sharp corners, rough spots and similar defects that could possibly damage items placed in the adapter or create a hazard to operating personnel. All bends of a major character shall be made with dies or fixtures to insure uniformity of size and shape. Adapters shall be free from distortion that will affect assembly of the unit load.

* 3.5 Preproduction Sample. Unless otherwise specified, as soon as practicable after award of contract and prior to submission of any adapter for acceptance, the contractor shall furnish a preproduction sample for determination of conformance to this specification. Examinations and tests shall be specified herein, see paragraph 4.4.1. Any changes or deviations from the preproduction sample will be subject to the approval of the contracting officer. The preproduction sample may not be included in the quantity to the shipped in the final shipment.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for Inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 Quality conformance inspection. The quality conformance inspection shall consist of:

(a) Qualification of weld process or machine (4.2.1)
(b) Production control tests (4.4.2, 4.5.1)
(c) Visual and dimensional examination (4.3.2, 4.5.2.3)
(d) Examination for preparation for delivery (4.5.2.1, and 4.5.2.4)

* 4.2.1 Qualification of weld process or machine. The weld process or machine shall be qualified under the requirements outlined in 3.3.1.1, 3.3.1.2 and 3.3.1.3.

4.3 Sampling

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4.3.1 Sampling for Production Control Tests. Sampling for production control tests shall consist of one production control coupon taken at the beginning and end of each shift and whenever the electrodes are changed or dressed, or when any changes are made in the control of the machine. Additional samples may be taken by the contractor during the shift to insure against rejection of produced adapters.

4.3.1.1 Preparation of Shear Test Welds. Five shear test specimens shall be selected at random from multiple welding coupons containing not less than eight consecutive welds. The coupons shall be of the same material and gage combinations which are to be used in production of the pallet adapters and shall be welded as required in production. The welding maching settings for these coupons shall be recorded.

4.3.1.2 Preparation of Tensile Test Welds. Five tensile test specimens shall be selected. The specimens shall be of the same material and gage combination which are to be used in the production of the pallet adapters and shall be welded as required in production. The welding machine settings for those specimens shall be recorded.

4.3.2 Sampling for Lot Acceptance. The following lot acceptance tests shall be performed in accordance with MIL-STD-105.

* (a) Visual and Dimensional Examination (4.5.2.1). Inspection level II, acceptable quality levels (AQL's) 1.0 percent for major defects and 2.5 percent for minor defects. In regard to critical defect class, for inspection lot sizes of 150 or less, use 100 percent of the lot. For inspection lot sizes of 151 to 3200, use sample of 150 units, acceptance number zero. For inspection lot sizes greater than 3200, use the sample size as indicated in MIL-STD-105 acceptance number zero.

(b) Protective coating (4.5.2.2) inspection level (S-3) AQL 1.0 percent.

(c) Preparation for delivery (4.5.2.1 and 4.5.2.4) Inspection Levels S-2 acceptance number zero.

* 4.4 Classification of Tests. The inspection and testing of pallet adapters shall be classified as follows:

(a) Preproduction tests

(b) Production Control tests

(c) Lot acceptance tests.

* 4.4.1 Preproduction Test. Examination and testing of preproduction sample shall be made for all applicable provisions of this specification. These examinations and tests shall be performed in the presence of a Government inspector prior to any acceptance by the Government and need not be repeated during the life of the contract unless a change is made in design, material, or method of construction.

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* 4.4.2 Production Control Test. Production control specimens shall be subjected to the production control tests of 4.5.1. Acceptance shall be based on no defects in a specimen. Failure of any specimen to meet these requirements will result in rejection of the adapters produced since the last satisfactory specimens. In case of failure, the welding method or equipment shall be modified and the test repeated.

* 4.4.3 Lot Acceptance Tests. These tests shall be accomplished on pallet adapters of each type to be submitted to the Government for acceptance under contract as specified in 4.5.2. Inspection and test procedures shall be submitted for Government approval prior to the commencement of production.

4.5 Inspection

4.5.1 Production Control Tests

* 4.5.1.1 Shear Test. The weld specimens shall be loaded to place the welds in shear. Each weld shall have a minimum shear value in pounds as specified in ASTM A 185 by the formula shown in 4.5.1.2 unless otherwise specified in applicable drawings. Welds tested shall meet the requirements of 3.3.1.2.

* 4.5.1.2 Shear value. Minimum average shear value in pounds force shall not be less than 35,000 multiplied by the area of the larger wire in square inches where the smaller wire is not less than size W 1.5 and has an area of 40 percent or more of the area of the larger wire.

4.5.1.3 Tension Test The weld specimens shall be loaded to place the welds in tension and to subject the weld to a force equal to 80 percent of the tensile strength of the wires joined. Welds tested shall meet the requirements of 3.3.1.3.

* 4.5.2 Quality Conformance Inspection.

* 4.5.2.1 Examinations. The examinations specified in 4.3.1 and 4.5.2.4 shall be accomplished on each pallet adapter submitted for acceptance under the contract to ensure that the pallet meets the requirements of 3.1.1, 3.2.1, and 3.4. Examinations shall either be performed by the manufacturer and witnessed by the Government Inspector, or they may be conducted by the Government Inspector.

* 4.5.2.2 Protective Coating. The pallet adapter shall be examined to determine conformance to 3.1.2.

* 4.5.2.3 Visual and dimensional examination. Each pallet adapter selected in accordance with 4.3.2 shall be visually inspected for completeness of manufacture, freedom from defects, proper identification, workmanship, and dimensions.

* 4.5.2.4 Preparation for delivery. Each fully prepared bundle of pallet adapters selected in accordnace with 4.3.2 shall be examined to determine conformance to section 5 of this specification.

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5. PACKAGING

5.1 Preservation and Packaging

5.1.1 Level C

5.1.1.1 Unit Packaging. Unless otherwise specified, the components of one pallet adapter shall be stacked together and secured with two 1/2 by 0.015 inch flat steel straps. Straps shall be in accordance with QQ-S-781 type 1 Finish A.

5.2 Packing

* 5.2.1 Level C. The pallet adapters shall be packed to afford protection against damage during direct shipment from the supply source to the first receiving activity. The packs shall comply with the following:

5.2.1.1 Palletization

* 5.2.1.1.1 Type 1. Twenty-four pallet adapters packed as described in 5.1.1.1 shall be stacked on a pallet fabricated in accordance with Drawing 1350881 with an "X" dimension of 40 inches and a "Y" dimension of 48 inches. Four 3/4 by 0.020 inch flat steel straps in accordance with QQ-S-781, type 1 finish A shall be used to secure unit packages to pallets. The palletizing pattern shall consist of six layers of four pallet adapters.

* 5.2.1.1.2 Type II. Eight pallet adapters packaged as described in 5.1.1.1 shall be stacked on a pallet fabricated in accordance with Drawing 1350881 with an "X" dimension of 40 inches and a "Y" dimension of 48 inches. Four 3/4 by 0.020 inch flat steel straps in accordance with QQ-S-781 type 1 finish A shall be used to secure unit packages to pallets.

* 5.2.1.1.3 Type III. Twelve pallet adapters packaged as described in 5.1.1.1 shall be stacked on a pallet fabricated in accordance with Drawing 1350881 with an "X" dimension of 40 inches and a "Y" dimension of 48 inches. Four 3/4 by 0.020 inch flat steel straps in accordance with QQ-S-781 type 1 finish A shall be used to secure unit packages to pallets. The palletizing pattern shall consist of three layers of four pallet adapters.

* 5.2.1.1.4 Type IV. Thirty four adapters, base, pallet shall be stacked on a pallet fabricated in accordance with Drawing 1350881 with an "X" dimension of 40 inches and a "Y" dimension of 48 inches. Four 3/4 by 0.020 flat steel straps in accordance with QQ-S-781, Type 1, finish A shall be used to secure unit packages to pallets. The palletizing pattern shall consist of seventeen layers of two adapters, base, pallet.

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* 5.2.1.1.5 Type V. Seventy-two adapters shall be stacked on a pallet fabricated in accordance with Drawing 1350881 with an "X" dimension of 47 5/8 inches and a "Y" dimension of 36 inches. Four 3/4 by 0.020 falt steel straps in accordance with QQ-S-781 type 1 finish A shall be used to secure unit packages to pallets. The palletizing pattern shall consist of eighteen layers of four adapters.

* 5.2.1.1.6 Type VI. Nine adapters shall be stacked on a pallet fabricated in accordance with Drawing 1350881 with an "X" dimension of 47 5/8 inches and a "Y" dimension of 36 inches. Four 3/4 by 0.020 flat steel straps in accordance with QQ-S-781 type 1, finish A shall be used to secure unit packages to pallets. The palletizing pattern shall consist of nine adapters stacked on each other.

* 5.2.1.1.7 Type VII. Ten adapters packed as described in 5.1.1.1 shall be stacked on a pallet fabricated in accordance with Drawing 1350881 with an "X" dimension of "40" and a "Y" dimension of 48 inches. Four 3/4 by 0.020 flat steel straps in accordance with QQ-S-781, type 1, finish A shall be used to secure unit packages to pallets. The palletizing pattern shall consist of five layers of two pallet adapters.

5.3 Marking. In addition to any special marking indicated on drawings or required by contract, unit packages and palletized loads shall be marked in accordance with MIL-STD-129.

* 6. NOTES AND CONCLUDING MATERIAL

6.1 Intended Use. The pallet adapter covered by this specification are intended for use in palletizing 5 inch projectiles and rocket heads, 6 inch 47 caliber projectiles, 8 inch 55 caliber projectiles, 4.4 inch chaff cartridge containers, bomb fin assembly MK 15 and bomb fin assembly MK 82.

* 6.2 Ordering Data. Procurement documents should specify the following:

(a) Title, number and date of this specification

(b) Type of pallet adapter being purchased

(c) Applicable drawings and other documents

(d) Quantity

(e) Place of inspection

(f) Packaging and palletization

(g) Place of delivery

(h) Exceptions to this specification

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* 6.3 The margins of this specification are marked with an asterick to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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DOCUMENT IDENTIFIER AND TITLE MIL-A-23685B(OS) ADAPTERS, PALLET, STEEL FOR PROJECTILES, ROCKET HEADS, CARTRIDGE CONTAINERS AND BOMB FINS

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
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1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
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